MAMR Study using New STO Design and Medium Optimization

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Microwave-assisted magnetic recording (MAMR) is a promising candidate for next-generation high-density hard disk drives, but requires both advanced spin-torque oscillator (STO) design and optimized media. We address these challenges from both perspectives. For STOs, we propose a novel SIL-free dual field-generation-layer (Dual-FGL) configuration (NP-STO), in which two FGLs with opposite spin polarization mutually act as spin sources, enabling near-zero oscillation delay time and stable operation. In parallel, we employ a simulation-based optimization scheme that combines read/write simulations with machine learning to optimize medium design, achieving a 25% recording density gain under CMR situation, while further optimization is ongoing. These results indicate the potential of MAMR as a future high-density magnetic recording technology.

***Index Terms*—Microwave-assisted magnetic recording, MAMR, Spin torque oscillator, STO**

I. Introduction

MICROWAVE-ASSISTED MAGNETIC RECORDING (MAMR) is a promising candidate for next-generation high-capacity hard disk drives (HDDs)[1]-[2]. MAMR utilizes the well-known microwave-assisted switching effect and is expected to enable higher areal recording densities. However, it requires several advanced technologies, including frequency matching between the microwave field and the ferromagnetic resonance frequency of the recording medium, as well as stable oscillation of a spin-torque oscillator (STO)[3].

To address these challenges, we have recently focused on STO designs incorporating two oscillation layers, known as dual field-generation-layer STOs (Dual-FGL STOs)[3]-[8], to enhance oscillation stability. In this study, we propose a novel Dual-FGL STO design with near-ideal oscillation characteristics. In addition, we present recent progress in MAMR medium design and its recording properties, focusing on candidate structures identified through an optimization scheme combined with machine learning techniques.

II. STO design chllenges and Dual-FGL STO

In MAMR head design, we have previously identified three key factors: (i) the interaction between STO and write pole[9], (ii) the negative FC effect[10], and (iii) the STO vertical field effect[11]. The Dual-FGL STO proposed by Toshiba is a promising candidate to meet these requirements.

As shown in figure 1(a), the conventional Dual-FGL STO (AIP-based) consists of two field-generation layers (FGLs) and one or two spin-injection layers (SILs). In this structure, the SIL switches during operation and injects spin current into the FGLs, similar to a conventional all-in-plane-type STO (AIP-STO). The two FGLs are coupled antiparallel via dipolar coupling and maintain this alignment even during oscillation in the 20–30 GHz frequency range. Furthermore, this dipolar coupling stabilizes the magnetization states of the FGLs, enabling oscillation with a large cone angle. Figure 1(b) shows the simulated oscillation behavior of conventional Dual-FGL STO designs. In this simulation, a full-writer model equipped with an STO in the write gap is employed, and the oscillation dynamics—including the interaction between the STO and the write pole—are analyzed by solving the Landau–Lifshitz–Gilbert (LLG) equation. Although the conventional Dual-FGL STO exhibits favorable characteristic in oscillation stability, it requires more than 0.2 ns to stabilize oscillation following magnetic pole reversal. This delay, defined as the oscillation delay time (ODT), significantly impacts recording performance. Figure 2 shows the effect of ODT on recording performance: the signal-to-noise ratio (SNR) degrades with increasing ODT, with a threshold of approximately 0.2 ns at a relative velocity of 22 m/s. Therefore, fast oscillation response is as critical as the other design factors of STOs. However, the AIP-based Dual-FGL STO cannot satisfy this requirement, mainly because SIL switching is essential for initiating FGL oscillation.

To address this issue, we propose a new Dual-FGL STO design, as shown in figure. 3[13]. This STO consists of only two FGLs with negative and positive spin polarization, respectively, and does not include an SIL. We refer to this configuration as an NP-STO. In the NP-STO, each FGL acts as a spin injection source for the other, enabling oscillation without an SIL. As a result, the ODT can be reduced to nearly zero by eliminating the SIL switching process. These results indicate that the NP-STO is a highly promising STO configuration for achieving high-density magnetic recording in MAMR..

III. Medium desing for MAMR

In MAMR, medium design is as important as STO design. We have experimentally observed assist-induced writability gain in actual recording processes[12]. However, further improvement in recording density is still required for practical applications. Medium design is highly challenging due to its multilayer structure with many parameters to be optimized, and its design strategy differs significantly from that of conventional media without microwave assistance.

To address this complexity, we employ an optimization scheme that combines read/write simulations with machine learning techniques. In this framework, we define BPI and TPI indices based on our design experience and optimize these metrics to maximize recording density. Figure 4 shows a comparison of recording performance between a medium candidate predicted by this optimization scheme and a production-based model. As a result, a recording density gain of approximately 25% has been achieved under conventional recording conditions (CMR). The optimization is ongoing. In the conference, we will report further improvements by MAMR through continued optimization and discuss its potential as a future high-density magnetic recording technology.

(a)

(b)

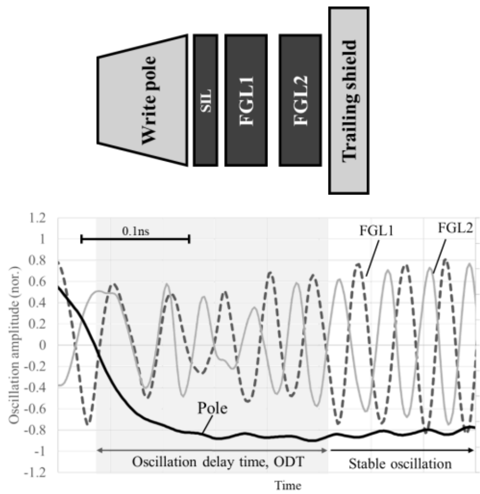

Fig.1 (a) configuration and (b) oscillation behavior of conventional Dual-FGL STO (AIP-based).

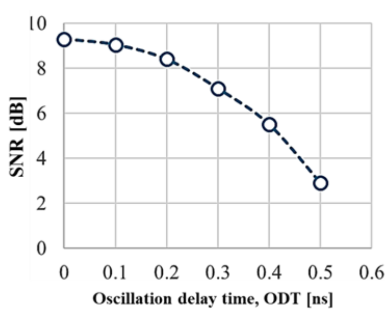

Fig.2 Oscillation delay time (ODT) impact to recording performance

(a)

(b)

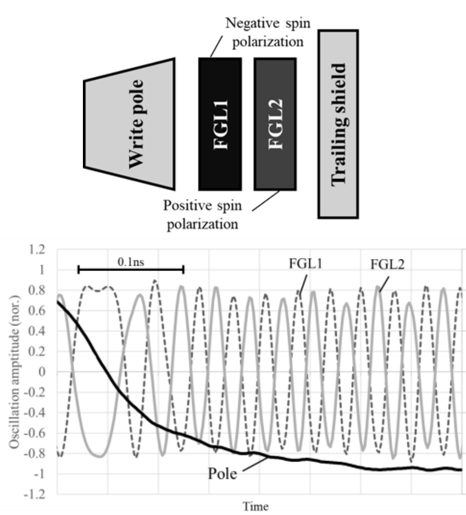

Fig.3 (a) configuration and (b) oscillation behavior of NP STO.

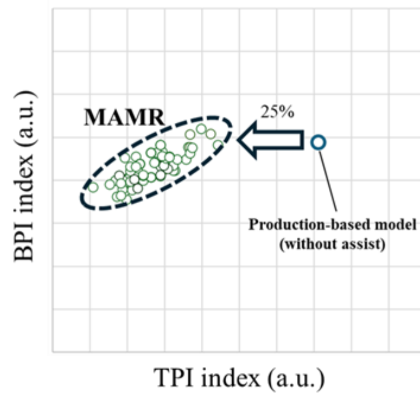

Fig.4 Comparison of recording density performance between production based-model and MAMR with .

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